Spatiotemporal modeling workflows with *tinyVAST*:
Multivariate vector autoregressive spatiotemporal (VAST) models

Sean Hardison, PhD

Rivanna Statistical Ecology Group LLC

In this lecture

A brief introduction to *tinyVAST* and some mechanics

tinyVAST syntax for fitting univariate index standardization models and *sdmTMB* comparisons

A bivariate VAST model of male and female hair crab CPUE

Specifying multiple distributions for partitions of data

A brief introduction to *tinyVAST*

Fits multivariate spatiotemporal models with lagged and simultaneous effects

Adopts arrow notation from structural equation path modeling to specify effects

- Extends to “arrow and lag” notation for specifying lagged effects

Allows for enormous model flexibility

- Spatial vector autoregression
- Empirical orthogonal function analysis
- Spatial factor analysis
- Univariate/multivariate species distribution models
- Smoothing spline response for covariates
- Multiple distributions/link functions for different partitions of data
- and many more…

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tinyVAST: R Package With an Expressive Interface to Specify Lagged and Simultaneous Effects in Multivariate Spatio-Temporal Models

James T. Thorson[1] | Sean C. Anderson[2] | Pamela Goddard[3] | Christopher N. Rooper[2]

[1]Resource Ecology and Fisheries Management, Alaska Fisheries Science Center, Seattle, Washington, USA | [2]Pacific Biological Station, Fisheries and Oceans Canada, Nanaimo, British Columbia, Canada | [3]Lynker Technologies, LLC, Under Contract to Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, Washington, USA

Correspondence: James T. Thorson (james.thorson@noaa.gov)

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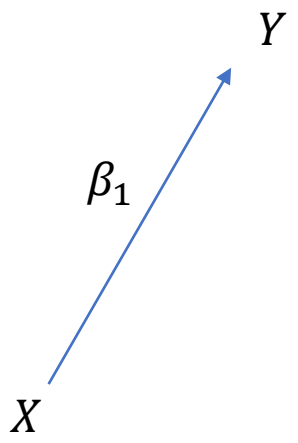
A brief introduction to *tinyVAST*

tinyVAST models fit to 4 components

1. **Spatial interactions among variables**
   - Interactions among variables within a site; do not change over time.
   - Spatial random effects
2. **Temporal interactions among variables**
   - Interactions among variables that apply equally everywhere; change over time but not location.
   - Simultaneous and lagged effects
3. **Spatiotemporal interactions among variables**
   - Simultaneous and lagged temporal interactions that vary by location
   - Spatiotemporal random effects
4. **Generalized additive model**
   - Generic *mgcv*-style formula interface for everything else!

$$\text{linear predictor} = \text{space term} + \text{time term} + \text{spacetime term} + \text{GAM}$$

Spatial interactions and arrow notation



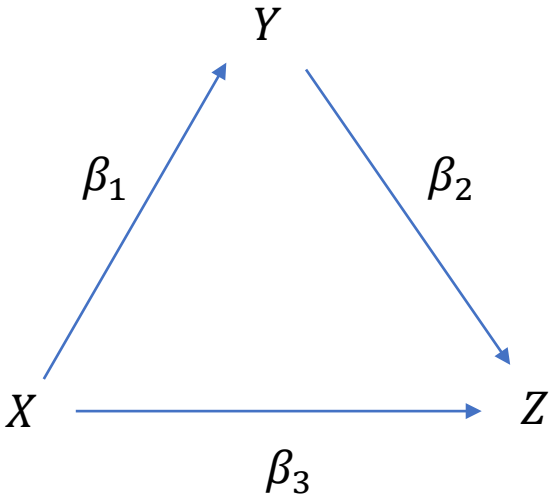
$$X \sim \text{Normal}(\mu_x, \sigma_X^2)$$
$$Y \sim \text{Normal}(\mu_Y + \beta_1 X, \sigma_Y^2)$$

Unidirectional arrow specifies an effect (slope)

$X \rightarrow Y$, beta1
$X \leftrightarrow X$, sigmaX
$Y \leftrightarrow Y$, sigmaY

Bidirectional arrow specifies variance or covariance

Path diagram



Linear model

$$X \sim \text{Normal}(\mu_x, \sigma_X^2)$$
$$Y \sim \text{Normal}(\mu_Y + \beta_1 X, \sigma_Y^2)$$
$$Z \sim \text{Normal}(\mu_Z + \beta_2 Y + \beta_3 X, \sigma_Z^2)$$

Arrow notation

$X \rightarrow Y$, beta1
$Y \rightarrow Z$, beta2
$X \rightarrow Z$, beta3
$X \leftrightarrow X$, sigmaX
$Y \leftrightarrow Y$, sigmaY
$Z \leftrightarrow Z$, sigmaZ

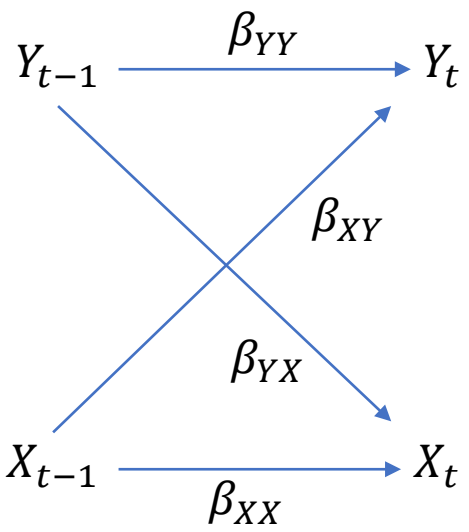
Don't need to be specified explicitly

Temporal and spatiotemporal interactions and arrow and lag notation

Two univariate time series X and Y have a cross-lagged interaction -- **bivariate vector autoregression**

- X at time t depends on X_{t-1} and Y_{t-1}, and equivalently for Y
- Can be interpreted as a **dynamic structural equation model (DSEM)** when no space term

An arrow and lag path diagram for this model



when specified in tinyVAST

$$X \rightarrow X, 1, \text{betaxx}, 0.1$$
$$X \rightarrow Y, 1, \text{betaxy}, 0.1$$
$$Y \rightarrow Y, 1, \text{betayy}, 0.1$$
$$Y \rightarrow X, 1, \text{betayx}, 0.1$$

Simultaneous or lagged effect

lag

parameter name

starting value

The statistical approach of *tinyVAST*

Random effects specified using GMRF precision matrices, as *sdmTMB* does for spatial and spatiotemporal random fields

The same models fitted in *sdmTMB* may be fitted in *tinyVAST*

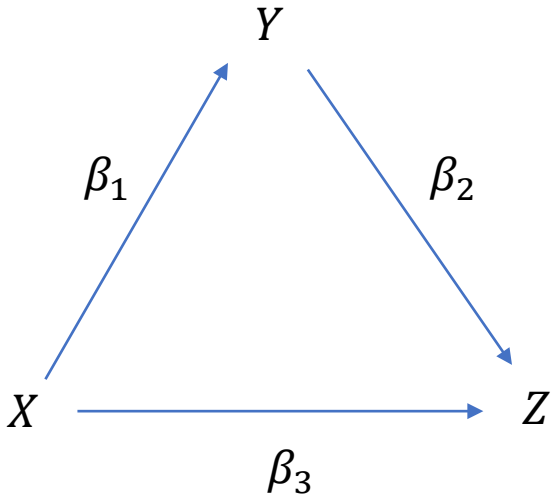
tinyVAST extend *sdmTMB* functionality by allowing for complex structural models to be integrated with spatial and spatiotemporal random fields

$$\text{linear predictor} = \underbrace{\text{space term} + \text{time term} + \text{spacetime term}}_{\text{Random effects using GMRFs}} + \text{GAM}$$

Spatial interactions as random effects

Sparse precision matrix $\boldsymbol{Q}_{spaceterm}$ assembled from matrices implied by the structural equation model

1. Path coefficients ($\boldsymbol{P}$)
2. Covariances ($\boldsymbol{\Gamma}$)



$\boldsymbol{Q}_{spaceterm}$ is then combined with the precision matrix for the spatial random field $\boldsymbol{Q}_{domain}$

$$\mathrm{vec}(\boldsymbol{\Omega}) = \mathrm{MVN}(\boldsymbol{0}, \boldsymbol{Q}_{spaceterm}^{-1} \otimes \boldsymbol{Q}_{domain}^{-1})$$

- Relationships between variables are captured by the structural model ($\boldsymbol{Q}_{spaceterm}$)
- Latent spatial relationships are captured by the spatial GMRF ($\boldsymbol{Q}_{domain}$)

if only the spatial term is specified, then the expected value for sample *i* and variable *c* is

$$p_i = \boldsymbol{A}_i \boldsymbol{\Omega}_{c(i)}$$

where $\boldsymbol{A}$ is a projector matrix linking mesh node to sampling location

Spatiotemporal interactions as random effects

Similar approach for spatiotemporal interaction term

Sparse precision matrix $\boldsymbol{Q}_{spacetime}$ assembled from matrices implied by the **dynamic structural equation model**

$\boldsymbol{Q}_{spacetime}$ is then combined with the precision matrix for the spatial random field $\boldsymbol{Q}_{domain}$

$$\mathrm{vec}(\mathbf{E}) = \mathrm{MVN}(\mathbf{0}, \boldsymbol{Q}_{spacetime}^{-1} \otimes \boldsymbol{Q}_{domain}^{-1})$$

$\boldsymbol{Q}_{spacetime}^{-1} \otimes \boldsymbol{Q}_{domain}^{-1}$ is a separable precision matrix for the sites × variables × times 3D array $\mathbf{E}$

if spatial and spatiotemporal random effects are specified, the linear predictor $p_{i,t}$ is then

$$p_{i,t} = \boldsymbol{A}_i \boldsymbol{\Omega}_{c(i)} + \boldsymbol{A}_i \mathbf{E}_{c(i),t(i)}$$

Estimating the model

Each of the random components of the *tinyVAST* model is associated with a sparse precision matrix

- Did not touch on temporal or GAM precision matrices but concept is similar

The marginal log-likelihood for the data and random effects is approximated using the Laplace method

Likelihood is optimized using gradients calculated via automatic differentiation in TMB

Mapping *tinyVAST* to univariate *sdmTMB* models

Add a categorical variable "var" to `data.frame` as column

- *tinyVAST* works in long format -- "var" specifies groupings (e.g., multiple species)
- Still need this for a univariate model

We'll use same mesh from sdmTMB for speed

Specifying a univariate index standardization model with **spatial random field** and **iid spatiotemporal random field**

- **`space_term`** specifies spatial random field only, no spatial interactions via SEM
- **`spacetime_term`** specifies spatiotemporal field only
  - 0 indicates no lag

```
# spatial random field
space_term <- "
  hair_crab <-> hair_crab, sd_space
"

spacetime_term <- "
  hair_crab <-> hair_crab, 0, sd_spacetime
"

m1_tv <- tinyVAST(
  space_term = space_term,
  spacetime_term = spacetime_term,
  data = hair_cpue,
  formula = cpue ~ 0 + as.factor(year),
  spatial_domain = tinyVAST_mesh,
  family = tweedie(),
  time_column = "year",
  space_columns = c("longitude","latitude"),
  variable_column = "var",
  control = tinyVASTcontrol(
    use_anisotropy = TRUE,
    newton_loops = 1 # sdmTMB sets newton_loo
sanity check
  )
)
```

Mapping *tinyVAST* to univariate *sdmTMB* models

`sanity_tv()` – still in Jim Thorson's dev branch – use with caution

- **`R/helpers.R`**

tinyVAST outputs map to model m1.2 from *sdmTMB* lab as expected

- Slightly different syntax

```
# A tibble: 10 × 3
   tinyVAST_years sdmTMB_year  year
             <dbl>       <dbl> <int>
 1         -0.178      -0.178  2014
 2         -0.822      -0.822  2015
 3         -1.62       -1.62   2016
 4         -1.12       -1.12   2017
 5         -1.29       -1.29   2018
 6         -1.87       -1.87   2019
 7         -2.06       -2.06   2021
 8         -2.03       -2.03   2022
 9         -1.89       -1.89   2023
10         -1.38       -1.38   2024
```

A model with AR1 correlated spatiotemporal random fields

```
space_term <- "
  hair_crab <-> hair_crab, sd_space
"

spacetime_term_ar1 <- "
  hair_crab -> hair_crab, 1, rho
  hair_crab <-> hair_crab, 0, sd_spacetime
"
```

The field at time *t* is the field at *t-1* times a correlation coefficient, *rho*

tinyVAST for univariate index standardization

We specify a random walk for the spatiotemporal random field by fixing the correlation at 1

Parameter name must be NA

```
# Random walk------
spacetime_term_rw <- "
  hair_crab -> hair_crab, 1, NA, 1
"
```

AR1/random walk are useful for projections and interpolations

- AR1 spatiotemporal random fields are mean reverting – random walks are not

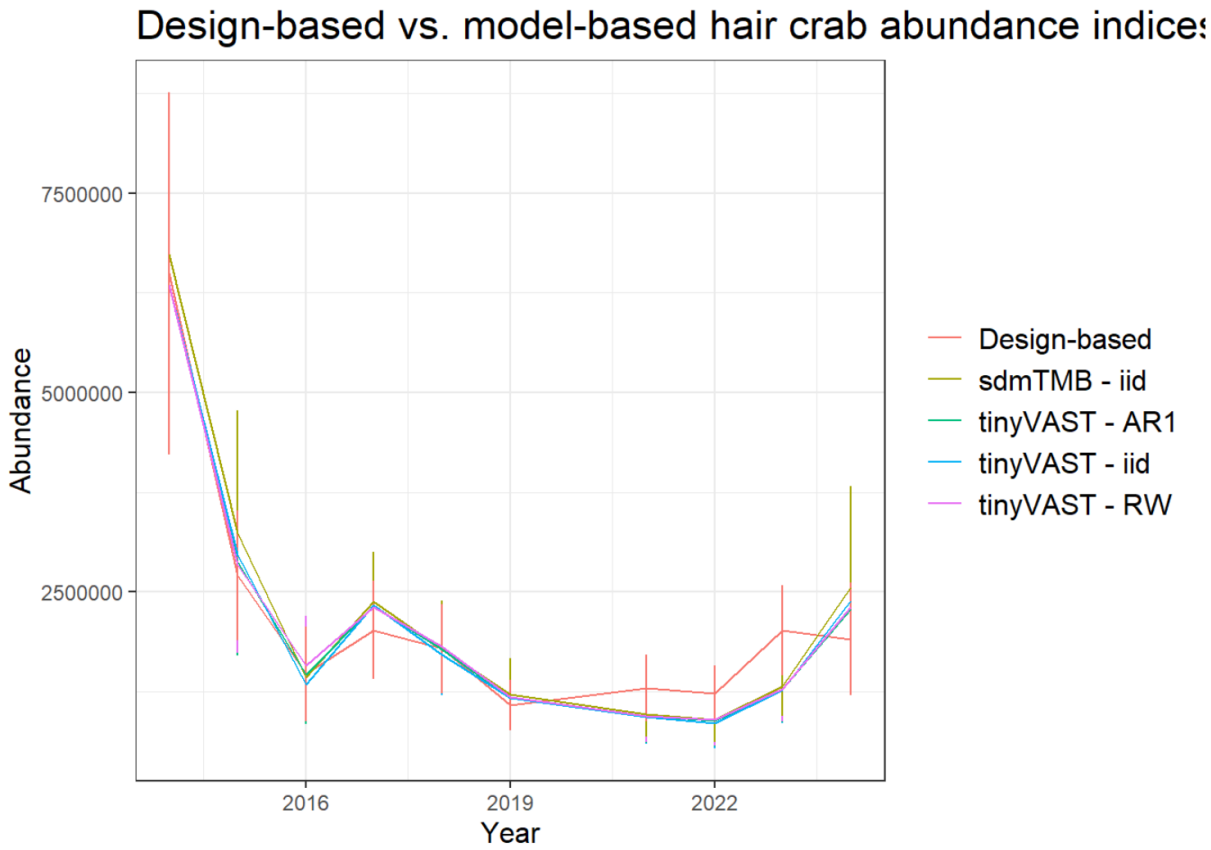
Marginal AIC suggests that the AR1/rw models are better fitting than the iid model

	df	AIC
m1_tv	17	3837.675
m1_tv_ar1	18	3828.573
m1_tv_rw	17	3831.275

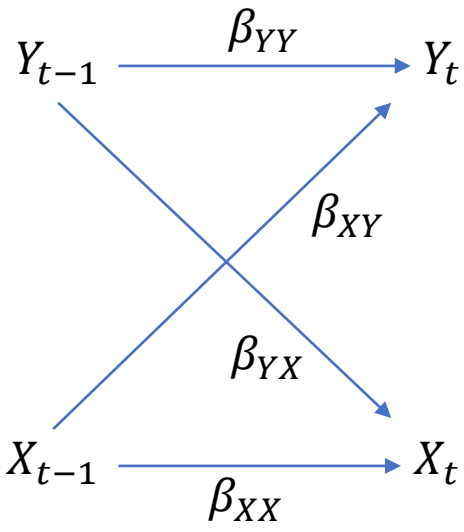
tinyVAST for univariate index standardization

The *tinyVAST* method for generating an aggregative index is **`integrate_output()`**

- Similar to *sdmTMB*, options for bias correction methods – applied in TMB
- New "type" argument to check out – type 1 is *sdmTMB* equivalent



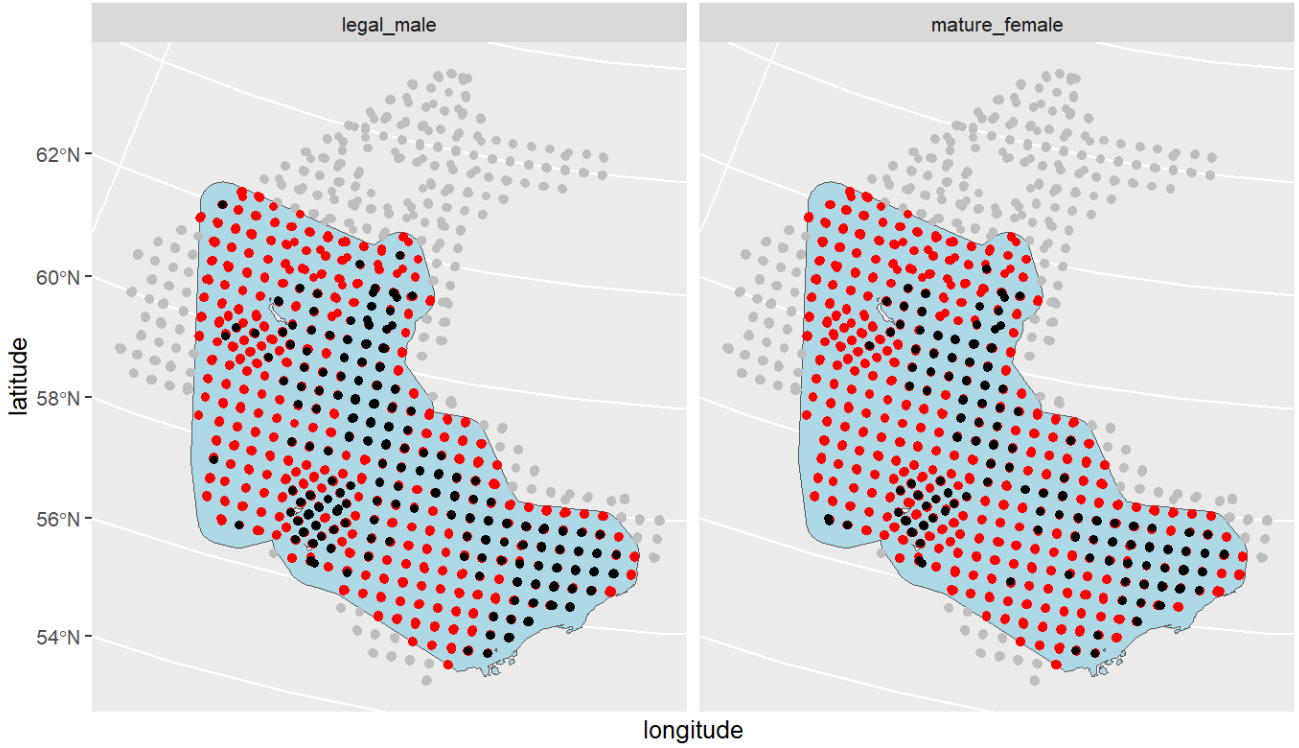
A bivariate VAST model



We specify a **bivariate vector autoregressive spatiotemporal model** for male and female hair crab CPUE

We first redefine the buffer zone for data to include in the model

- Now includes the region encompassing both legal male and mature female hair crab CPUE



A bivariate VAST model

Specifying spatiotemporal interactions

1. Autoregressive effects of year *t-1* on year *t*

2. Crossed effects of mature female CPUE at *t-1* on legal male CPUE at time *t*, and vice versa

3. Spatiotemporal random fields, *iid*, different for each sex category

Dropped spatial random field and anisotropy to aid convergence

```
spacetime_term_vast <- "
  # autoregressive effects
  legal_male -> legal_male, 1, b11
  mature_female -> mature_female, 1, b22

  # cross lagged effects
  mature_female -> legal_male, 1, b21
  legal_male -> mature_female, 1, b12

  # spatiotemporal variances (seperate)
  legal_male <-> legal_male, 0, sd_spacetime1
  mature_female <-> mature_female, 0, sd_spacetime2
"
```

A bivariate VAST model

Specifying spatiotemporal interactions

1. Autoregressive effects of year *t-1* on year *t*

2. Crossed effects of mature female CPUE at *t-1* on legal male CPUE at time *t*, and vice versa

3. Spatiotemporal random fields, *iid*, different for each sex category

```
# fit model
m2_tv <- tinyVAST(
          spacetime_term = spacetime_term_vast,
          data = hair_cpue_mf,
          time_column = "year",
          space_columns = c("longitude","latitude"),
          variable_column = "var",
          formula = cpue ~ 0 + var,
          spatial_domain = tinyVAST_mesh2,
          family = tweedie() )
```

Dropped spatial random field and anisotropy to aid convergence

“var” categorical predictor to estimate sex-specific intercepts

A bivariate VAST model: predictions and residuals

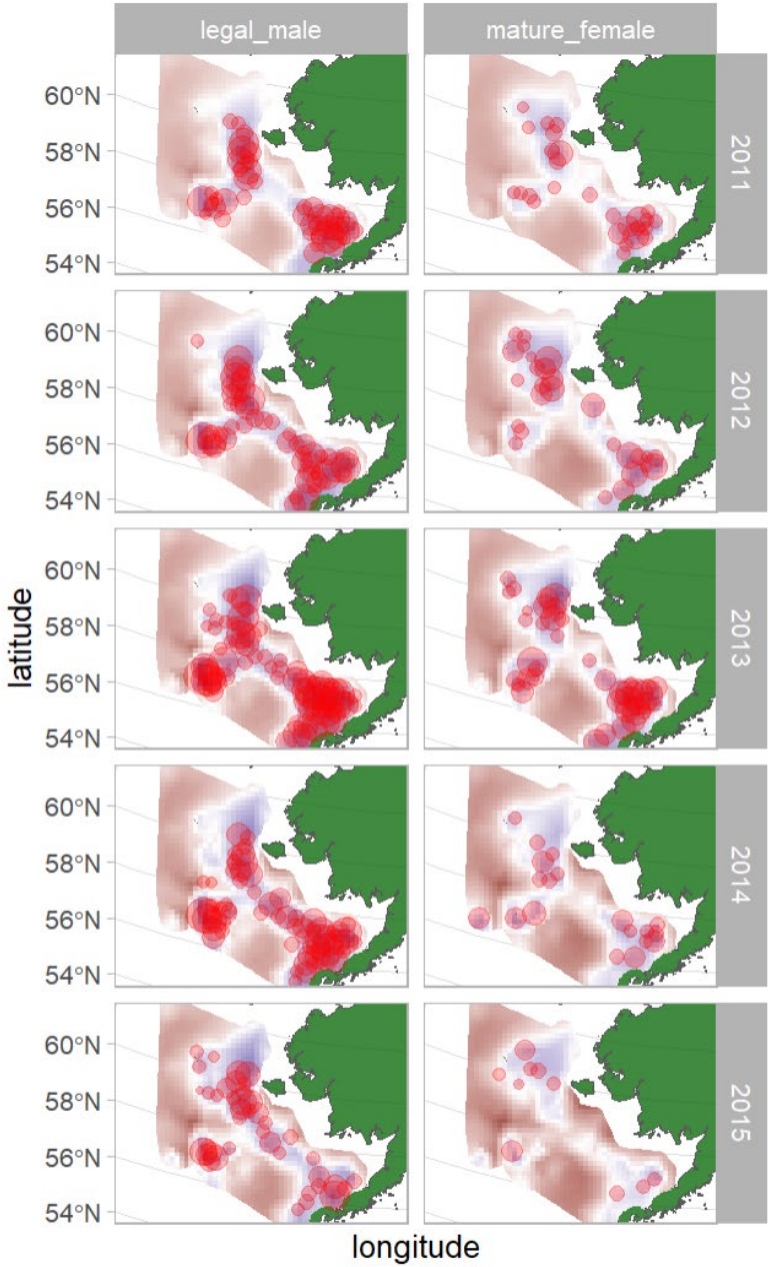
mgcv-style prediction interface instead of tidy *sdmTMB* style

Lots of options for prediction - see `?tinyVAST::predict.tinyVAST`

```
predict(
  object,
  newdata,
  remove_origdata = FALSE,
  what = c("mu_g", "p_g", "p1_g", "palpha1_g", "pgamma1_g", "pepsilon1_g", "pomega1_g",
    "pdelta1_g", "pxi1_g", "p2_g", "palpha2_g", "pgamma2_g", "pepsilon2_g", "pomega2_g",
    "pdelta2_g", "pxi2_g"),
  se.fit = FALSE,
  bias.correct = FALSE,
  ...
)
```

Residuals – options are DHARMa, deviance, and response

- DHARMa residuals for this VAST model sometimes work, sometimes break R



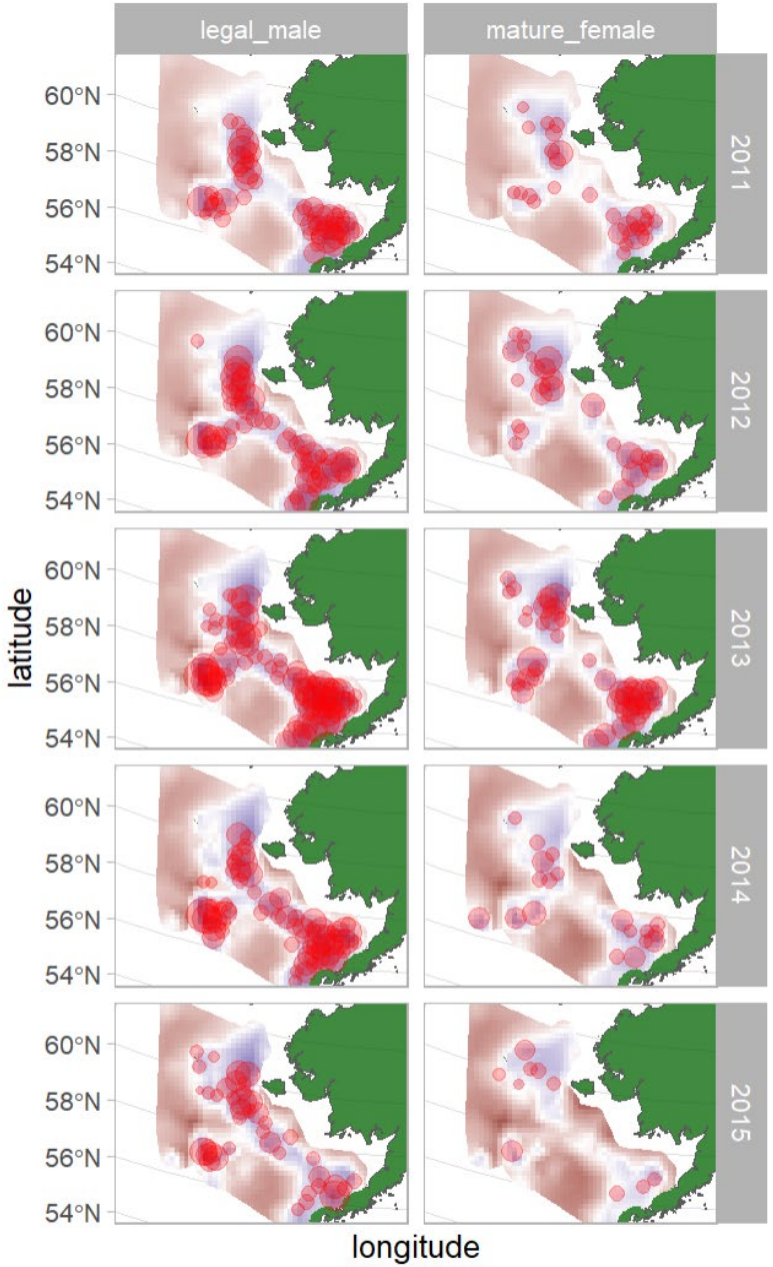
A bivariate VAST model: multiple distributions

We can specify different response distributions for different partitions of data

- Partitions specified by “var”
- A `distribution_column` argument is also required here – set to column with partitioned categorical predictor

```
fam_list <- list(
  legal_male = tweedie(),
  mature_female = tweedie()
)

m3_tv <- tinyVAST(
         spacetime_term = spacetime_term_vast,
         data = hair_cpue_mf,
         time_column = "year",
         space_columns = c("longitude","latitude"),
         variable_column = "var",
         distribution_column = "var",   # must be supplied with >1 family arg
         formula = cpue ~ 0 + var,
         spatial_domain = tinyVAST_mesh2,
         family = fam_list )
```



Tweedie parameters virtually identical between male and female hair crab

Spatiotemporal modeling lab 2

Please turn your scripts to `R/tinyVAST_lab_exercises.qmd`

Further reading

Thorson, James, and Kasper Kristensen. *Spatio-temporal models for ecologists*. Chapman and Hall/CRC, 2024.

Thorson, James T., et al. "tinyVAST: R package with an expressive interface to specify lagged and simultaneous effects in multivariate spatio-temporal models." *Global Ecology and Biogeography* 34.4 (2025): e70035.

Thorson, James T., et al. "Dynamic structural equation models synthesize ecosystem dynamics constrained by ecological mechanisms." *Methods in Ecology and Evolution* 15.4 (2024): 744-755.

Thorson, James T., et al. "Spatial factor analysis: a new tool for estimating joint species distributions and correlations in species range." *Methods in Ecology and Evolution* 6.6 (2015): 627-637.